

University of Stuttgart
Germany

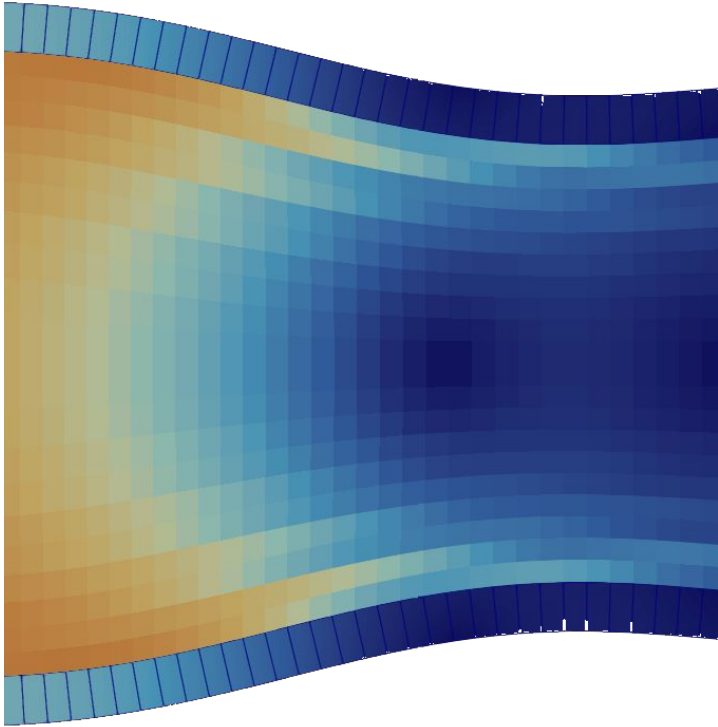
Fluid-Structure Interaction 2D Elastic Tube

Final Project
Numerical Simulation

**Benedikt Rösch
Julian Schmoll
Fabio Tucciarone**

Outline

Fluid-Structure Interaction

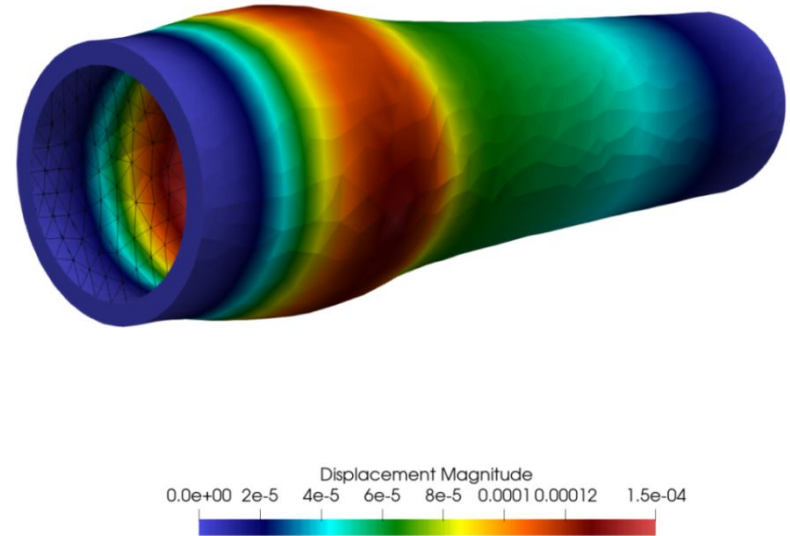


1. Motivation
2. Solver Adjustments
3. Coupling with preCICE
4. Structure Participant
5. Results
6. Validation
7. Outlook

Motivation

and 3D Elastic Tube example

- Learn how to use/configure established simulation tools
- Verify results of own solver
- Coupling of multiple Simulations
- Getting closer to a “*real world application*”
- Some nice pictures :)



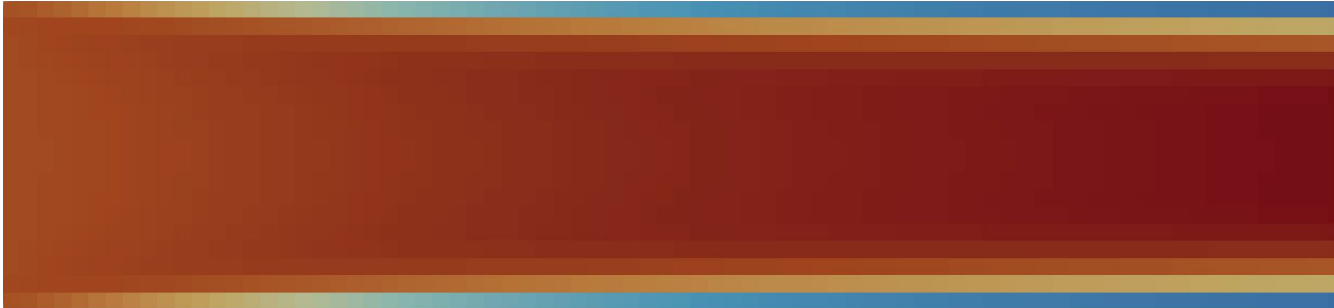
Figures: [F1]

Solver Adjustments

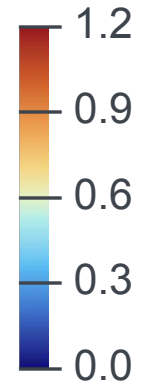
Channel Flow ($t=20s$)



Our Solver



OpenFOAM

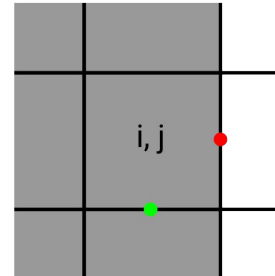
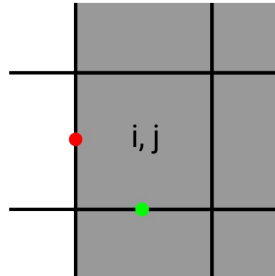
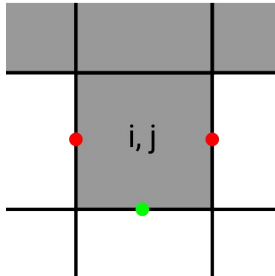
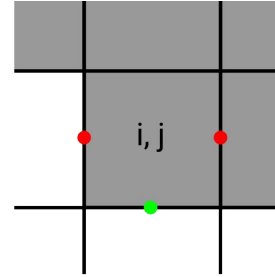
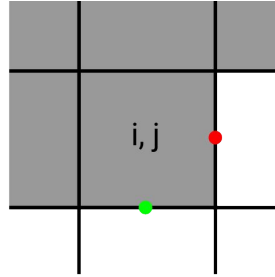
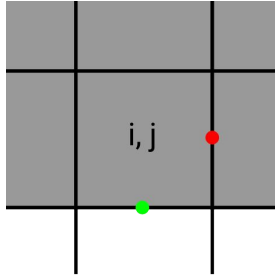


Velocity Magnitude

Solver Adjustments

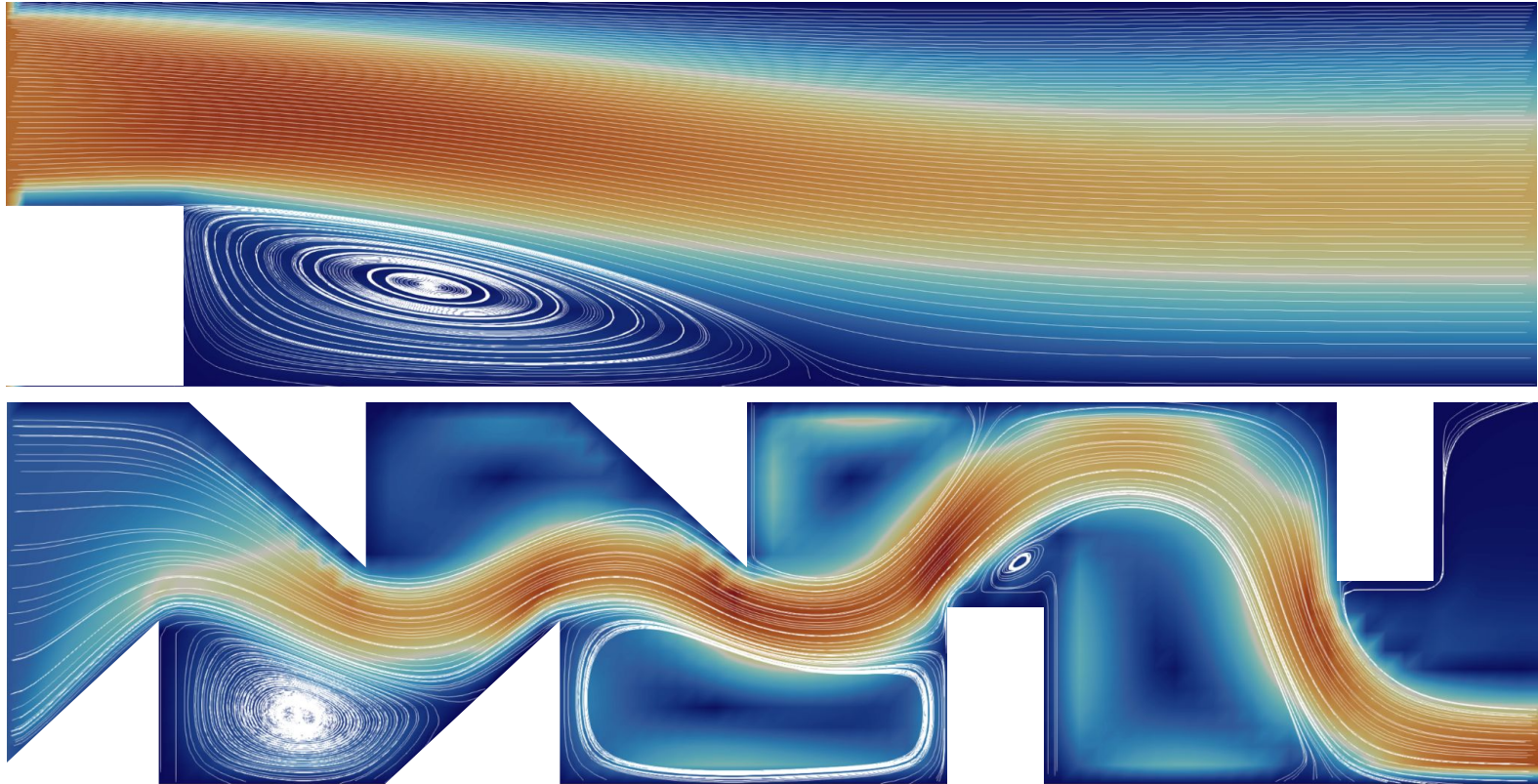
General Boundary Conditions

Top Boundary



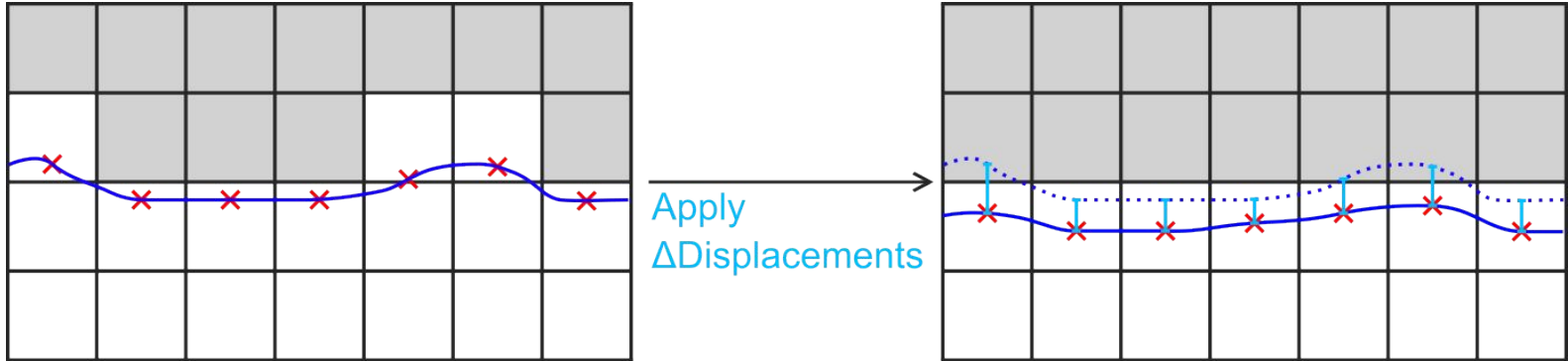
Solver Adjustments

General Boundary Conditions



Solver Adjustments

Example Boundary Geometry



- actual boundary
- stored values: `std::vector<double>`
- boundary cells: `Array2d<bool>`

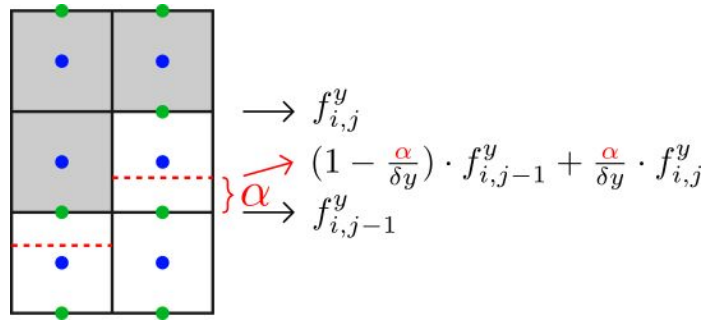
Solver Adjustments

Implementation of Forces

Only consider forces and displacements orthogonal to upper and lower boundary segments:

$$f^y = \int_{\Gamma_{\text{Body}}} \left(\frac{1}{\text{Re}} \nabla v - \begin{bmatrix} uv \\ v^2 + p \end{bmatrix} \right)^T \cdot \vec{n} \, dS \quad \vec{n} = \pm \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Interpolate forces on the actual boundary position:



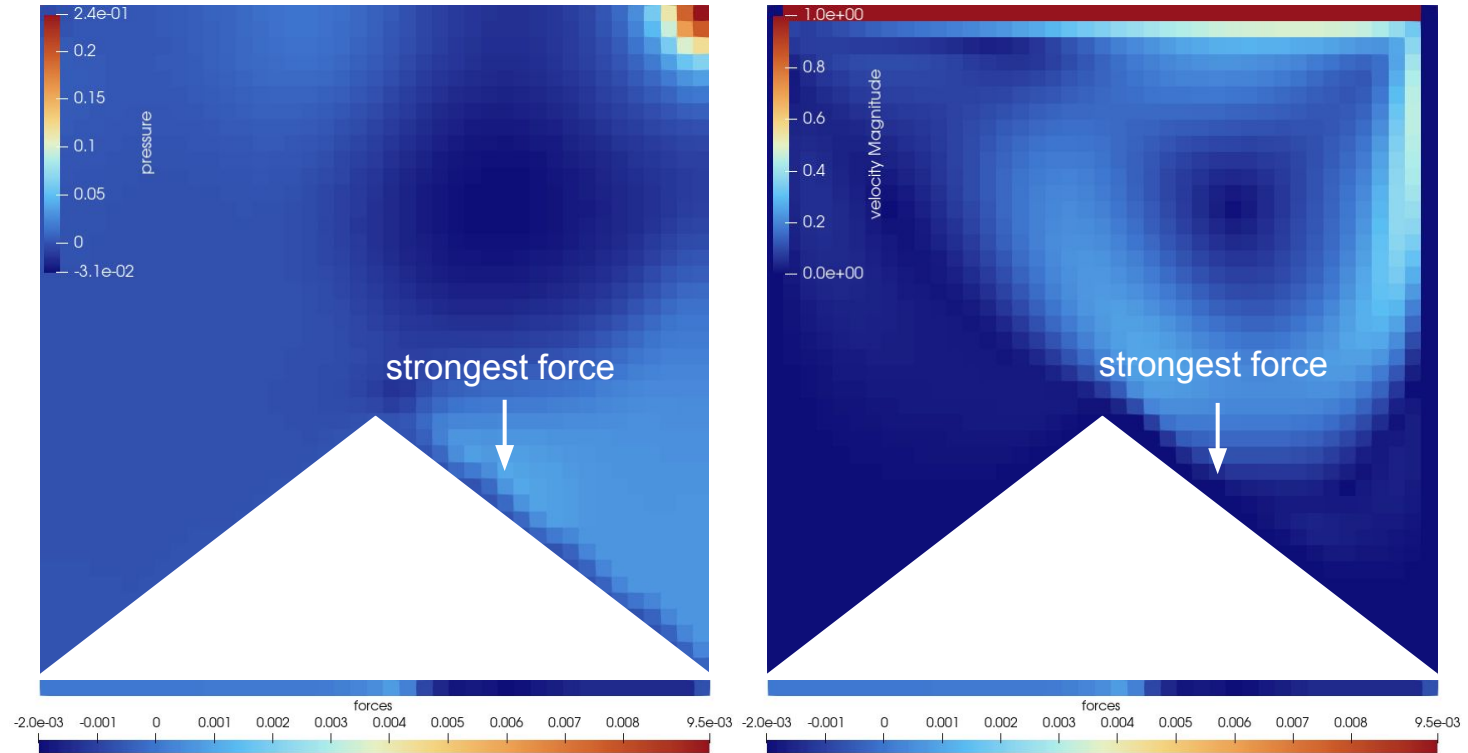
Solver Adjustments

Correction of velocities after modification of cells

```
new_fluid_cells_created := update_structure()  
if new_fluid_cells_created:  
    set_boundaries()  
    set_rhs()  
    set_preliminary_velocities()  
    q := solve_pressure(rhs)  
    update_velocities(q)
```

Solver Adjustments

Boundary and Forces - Plausibility Checks



Boundary and Forces - Plausibility Checks

- CalculiX's internal displacements (red):

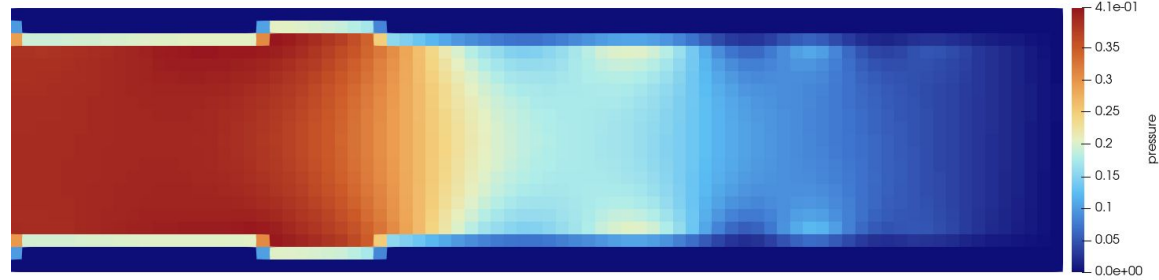


11

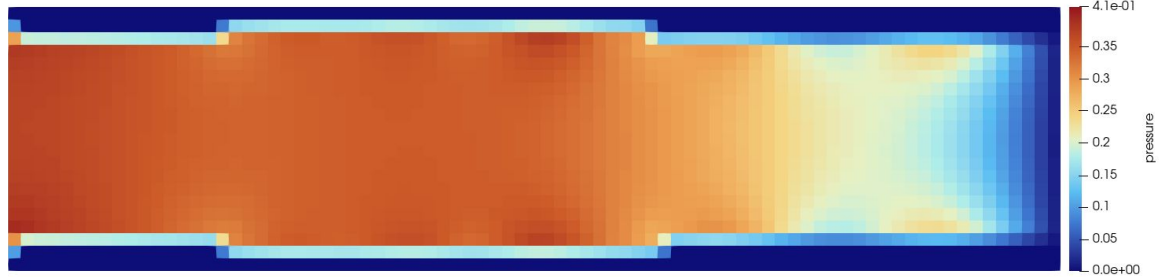
Solver Adjustments

Boundary and Forces - Plausibility Checks

t=0.50s

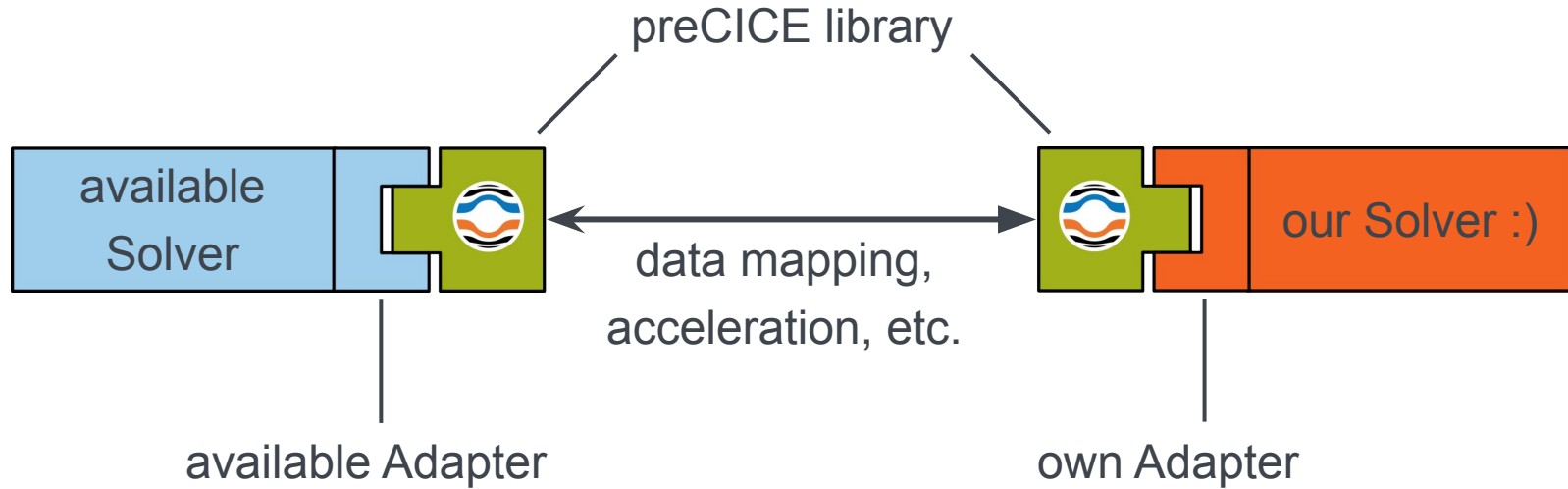


t=1.00s



Coupling with preCICE

Introduction

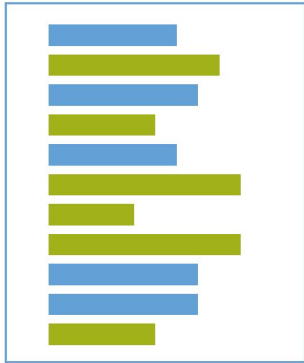


Sources: [1]; Figures: Adapted from [F2]

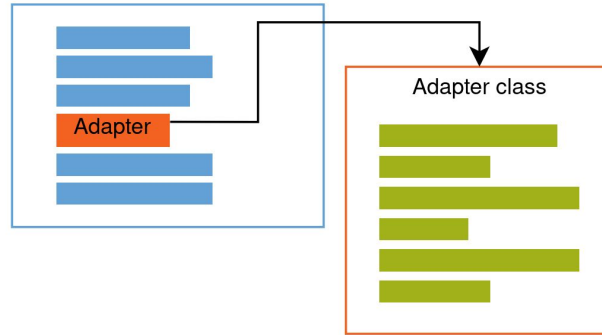
Coupling with preCICE

Adapter Architecture

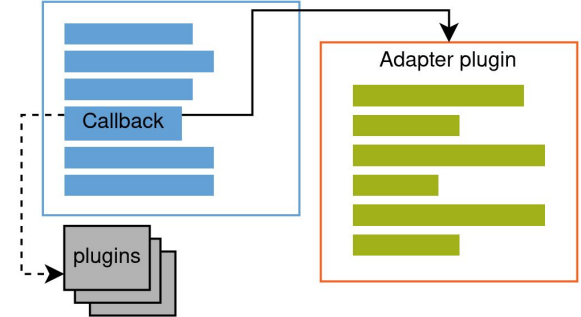
Direct modification:
solver + preCICE API calls



Adapter class:
solver calling adapter



Plugin:
solver with callback functionality



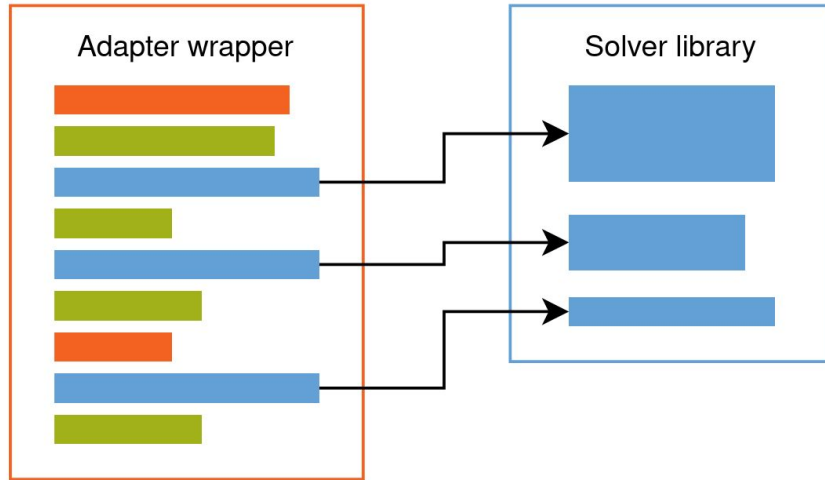
■ Solver code lines ■ Adapter code lines ■ preCICE API code lines

Figures: [F3]

Coupling with preCICE

Adapter Architecture

Wrapper:
adapter calls solver and preCICE



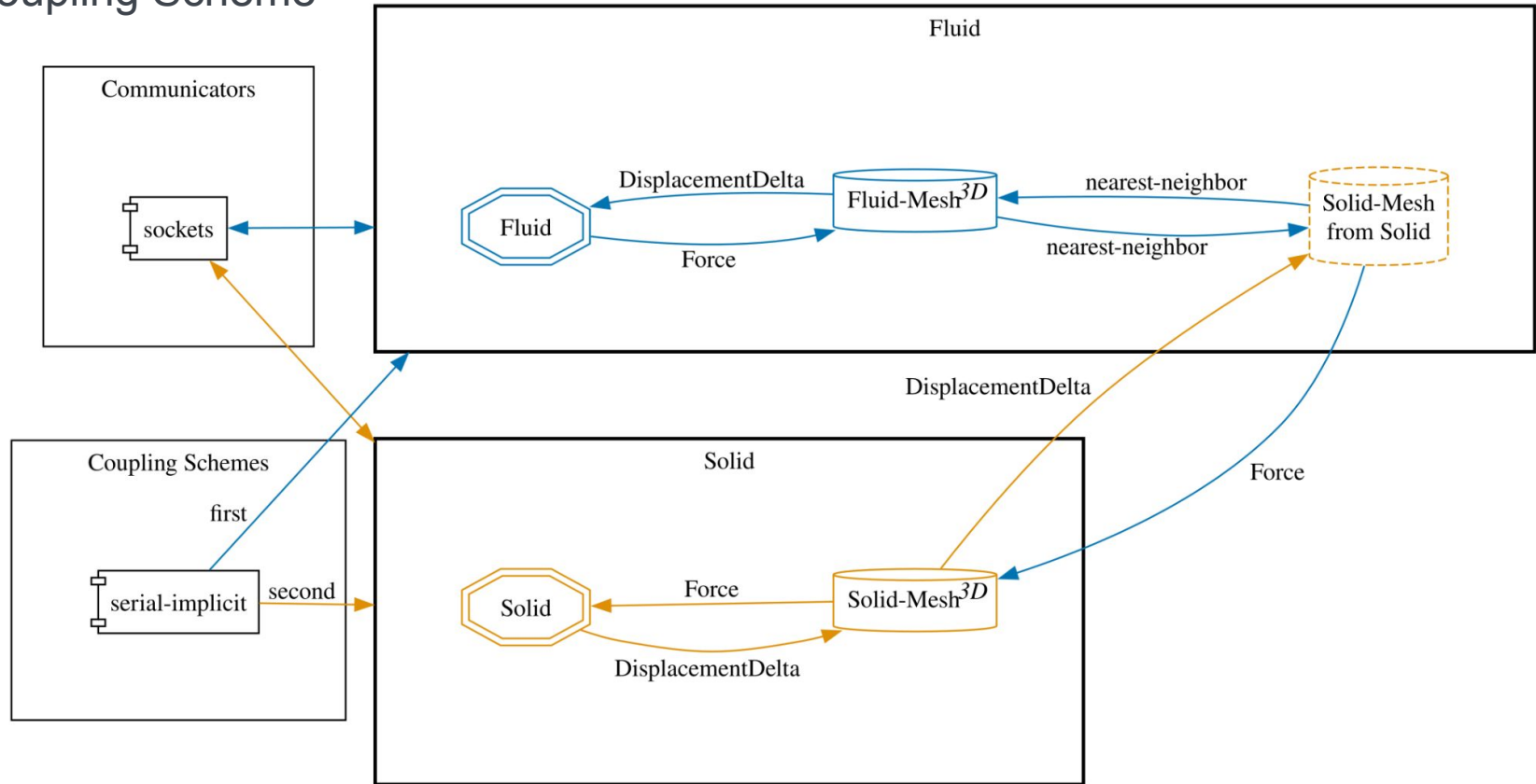
- Solver remains encapsulated and is built without preCICE
- Separate Binary for Adapter
- Needs restructuring of Solver Code to be called as API

■ Solver code lines ■ Adapter code lines ■ preCICE API code lines

Figures: [F3]

Coupling with preCICE

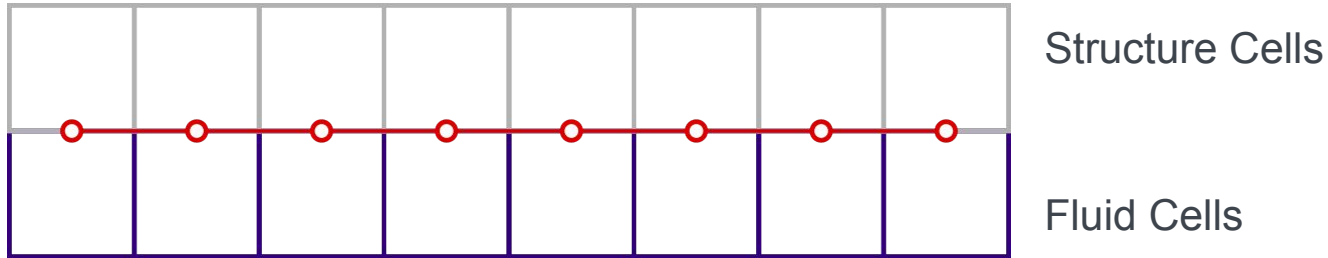
Coupling Scheme



Coupling with preCICE

Data Mapping

- Data in preCICE lives on vertices of the mesh, while the forces and displacements are calculated/applied for cell centers in [our solver](#).
- Our **preCICE Mesh** therefore consists of two lines of vertices (one at top of the fluid domain and one at bottom boundary) aligned like this:

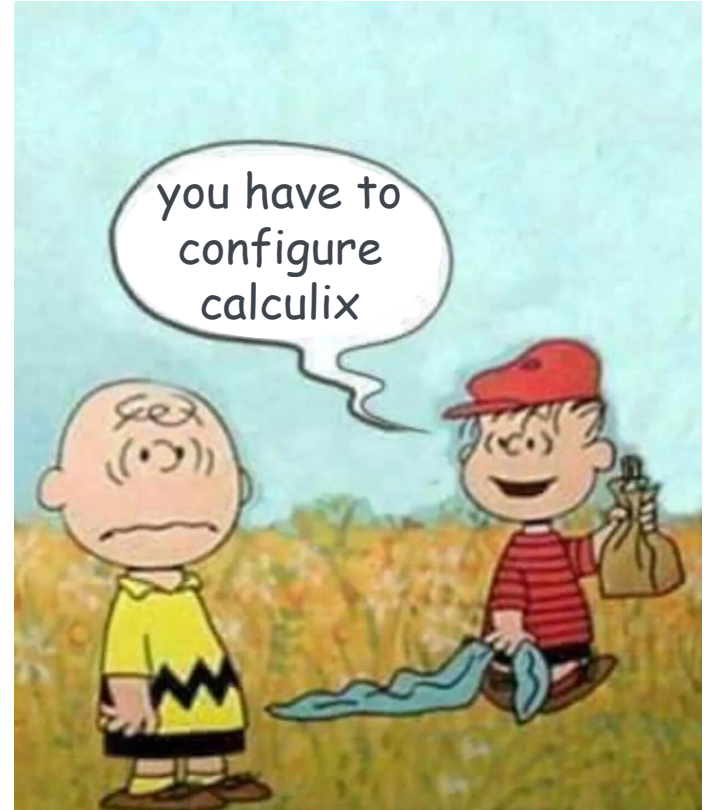


- Positions can exactly match the positions of Structure preCICE Mesh, nearest neighbor mapping is accurate

Structure Participant Overview

- Structure participant must calculate displacement based on material properties and forces
- We considered using FEniCS or CalculiX
- Adapters exist for both of them
- CalculiX seemed easier to run and more flexible
- New configuration files and mesh generation methods needed

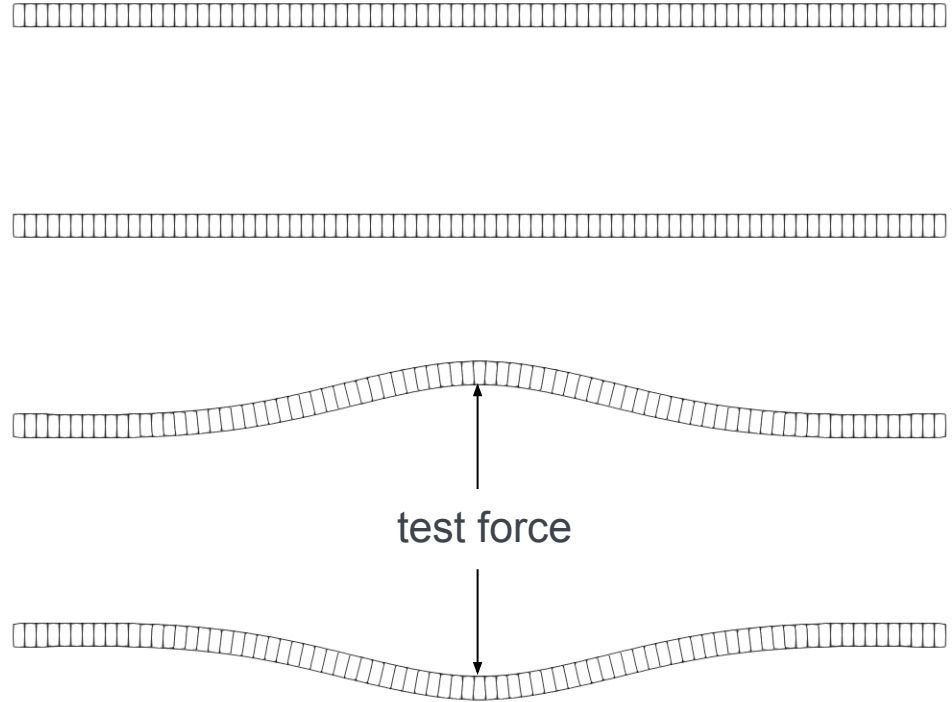
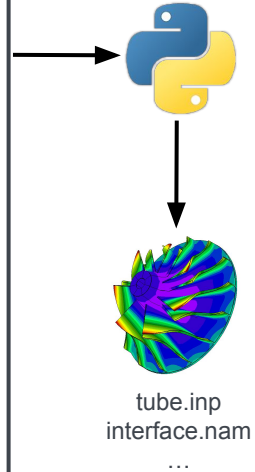
Figures: Adapted from [F4]



Structure Participant

CalculiX

```
geometry:  
  length: 8.0  
  height: 2.0  
  wall_thickness: 0.2  
  n_elements_length: 80  
  
material:  
  youngs_modulus: 300000  
  poissons_ratio: 0.3  
  density: 1200  
  
walls:  
  top:  
    active: true  
    offset: 0.2  
    fixed_start: true  
    fixed_end: true  
  bottom:  
    active: true  
    offset: 0.2  
    fixed_start: true  
    fixed_end: true
```



Figures: [F5], [F6]

Structure Participant

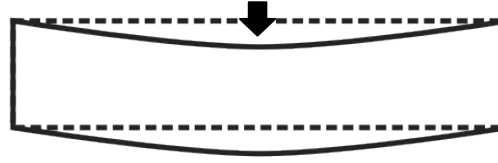
Material Properties

```
geometry:  
  length: 8.0  
  height: 2.0  
  wall_thickness: 0.2  
  n_elements_length: 80
```

```
material:  
  youngs_modulus: 300000  
  poissons_ratio: 0.3  
  density: 1200
```

```
walls:  
  top:  
    active: true  
    offset: 0.2  
    fixed_start: true  
    fixed_end: true  
  bottom:  
    active: true  
    offset: 0.2  
    fixed_start: true  
    fixed_end: true
```

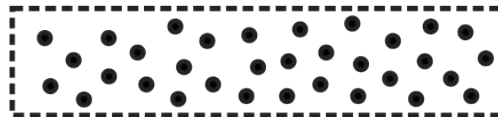
Young's Modulus
Stiffness of the Material



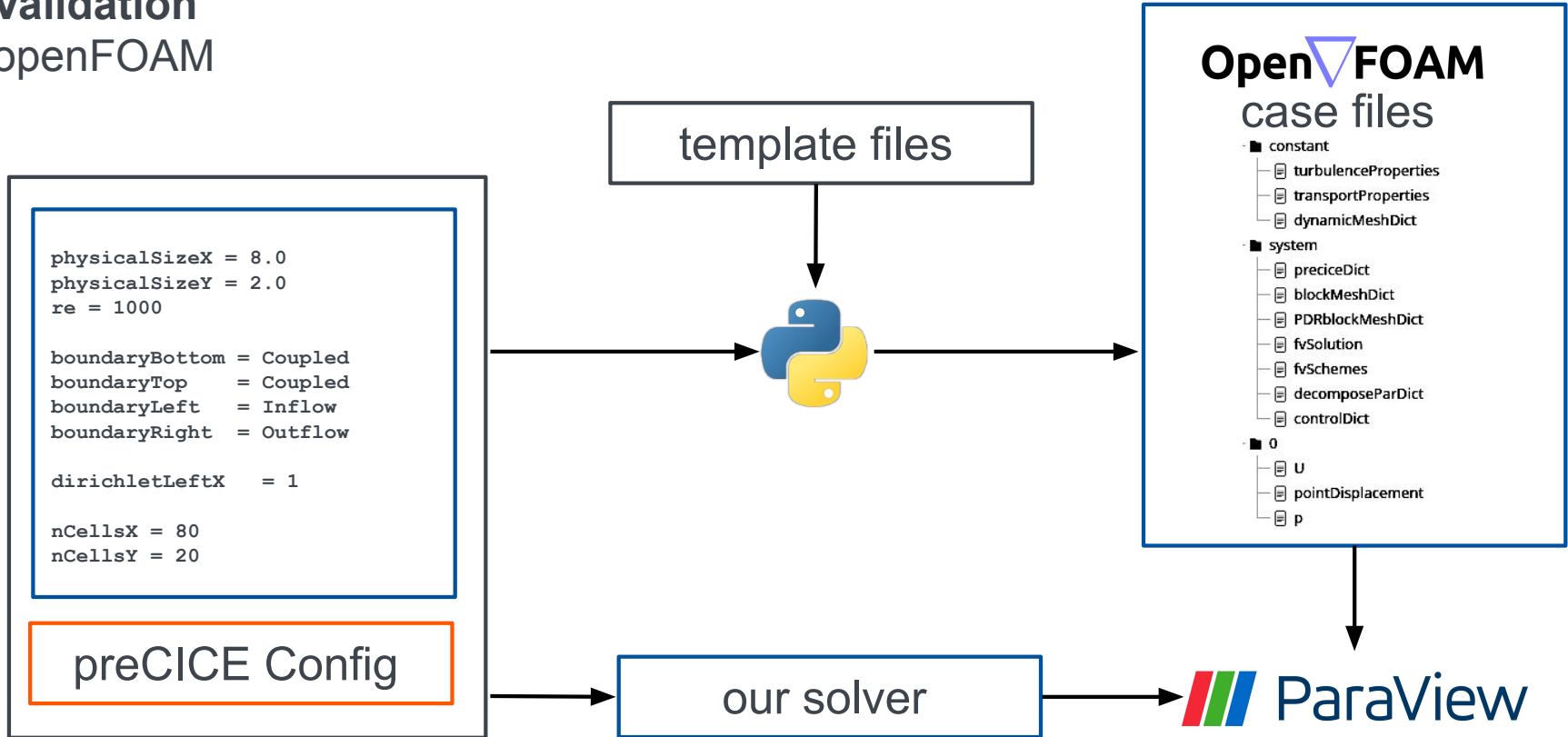
Poisson's Ratio
Deformation
perpendicular to load



Density
Mass per volume



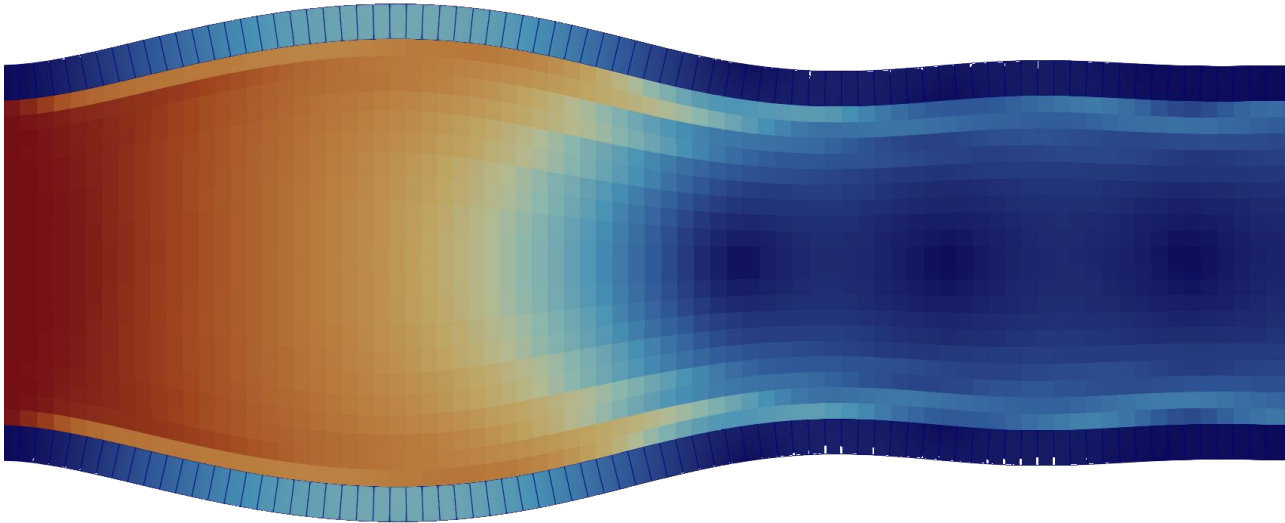
Validation openFOAM



Figures: [F5], [F7], [F8]

Validation

Elastic Tube



t: 1.0

Re: 1000

Field: 8.0x2.0

Youngs Modulus: 300000

Poissons Ratio: 0.3

Density: 1200

Results

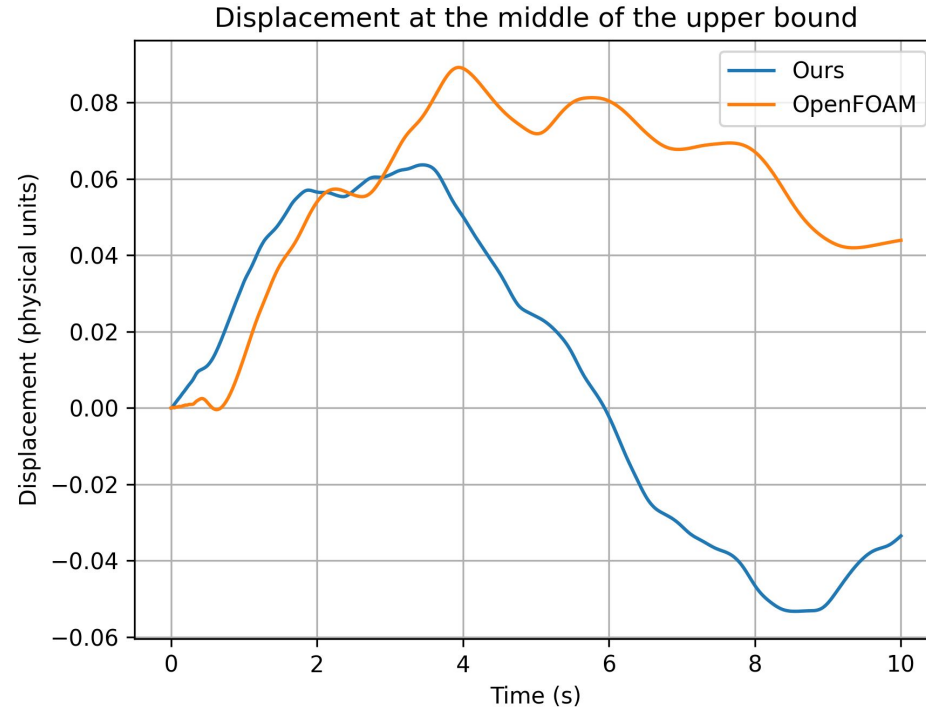
Elastic Tube



Real time, 10x enhanced displacement

Results

Elastic Tube



Outlook

Further Ideas and Improvements

- Validation
 - Validate coupled Simulation with OpenFOAM (Error plots)
 - Compare coupling results with CalculiX to FEniCS
- Improving adapter and solver
 - Adding second force component for more complex scenarios
 - Make it possible to define arbitrary structures
 - Parallelization currently buggy with adapter and forces
 - Implement Heat
 - Extend to 3D Domain

Sources

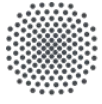
Figures

- [F1] <https://precice.org/images/tutorials-elastic-tube-3d-tube-result.png>
- [F2] <https://precice.org/images/lead-image-precice.svg>
- [F3] <https://precice.org/images/docs/couple-your-code/adaptersoftwareengineering/adaptersoftwareengineering-approaches.svg>
- [F4] <https://theawesomedaily.com/wp-content/uploads/2023/07/peanuts-7-1140x1420.jpeg>
- [F5] <https://www.python.org/static/img/python-logo-large.png>
- [F6] <https://upload.wikimedia.org/wikipedia/commons/3/34/Lav.png>
- [F7] <https://www.paraview.org/wp-content/uploads/2022/09/ParaView-Logo.svg>
- [F8] https://doc.openfoam.com/2306/img/openfoam_logo.jpg
- [F9] <https://www.watson.ch/imgdb/1277/Qx,B,0,0,355,400,333,166,133,66/7681925891976728>

Sources

Bibliography

- [1] G. Chourdakis et al., “preCICE v2: A sustainable and user-friendly coupling library,” Open Research Europe, vol. 2, no. 51, 2022, doi: 10.12688/openreseurope.14445.2.
- [2] G. Chourdakis et al., “OpenFOAM-preCICE: Coupling OpenFOAM with external solvers for multi-physics simulations,” OpenFOAM Journal, vol. 3, 2023, doi: 10.51560/ofj.v3.88.
- [3] B. Uekermann et al., “Official preCICE Adapters for Standard Open-Source Solvers,” in Proceedings of the 7th GACM Colloquium on Computational Mechanics for Young Scientists from Academia, Oct. 2017. doi: 10.18419/opus-9334



University of Stuttgart
Germany

Thank you!



Final Project Numerical Simulation

Benedikt Rösch	3870444
Julian Schmoll	3870431
Fabio Tucciarone	3589861